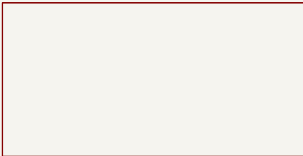


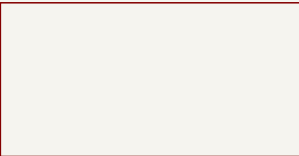
Data Center RDIMM DDR5 Tester

HyperRAM



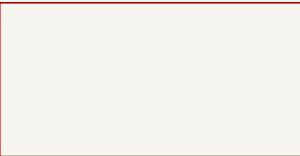
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DDR5



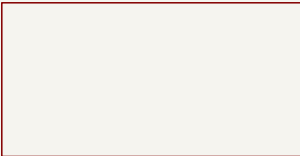
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Interfaces



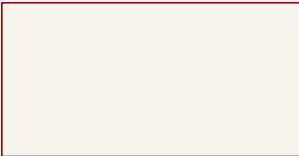
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Ethernet



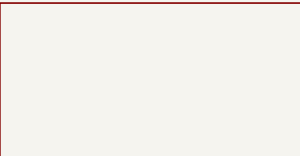
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Supply



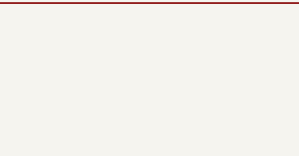
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Config SPI flash



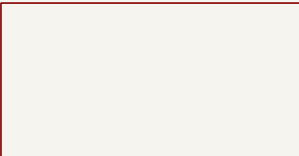
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FPGA power



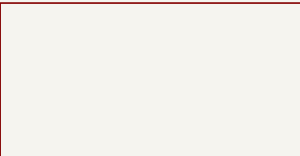
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FPGA banks 12-15



File: fpga-banks-12-15.kicad_sch

FPGA banks 16-34



File: fpga-banks-16-34.kicad_sch

Logo N2
oshw_logo

Logo N1
antmicro_logo



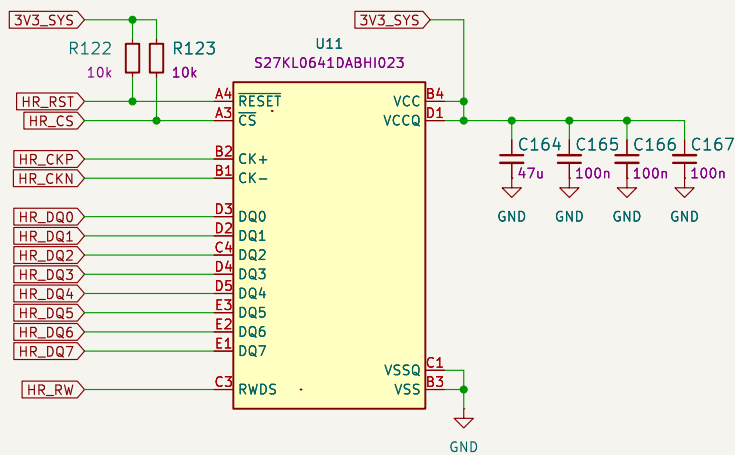
Antmicro Ltd
www.antmicro.com

Sheet: /
File: data-center-rdim--ddr5-tester.kicad_sch

Title: Data Center RDIMM DDR5 Tester

Size: A3	Date: 2023-07-27	Rev: 1.0.1:7e893
KiCad E.D.A. 8.0.8	Id: 1/10	

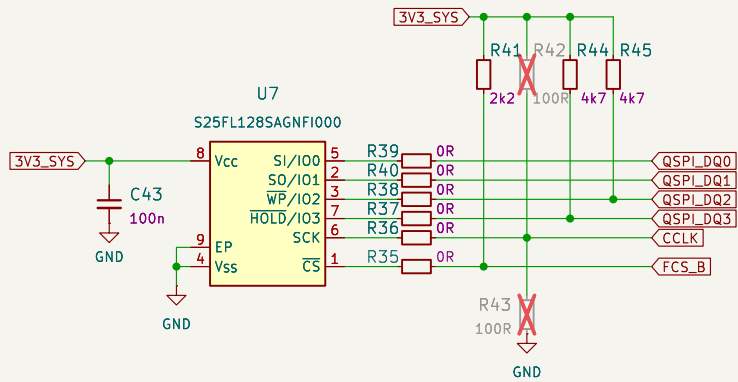
HyperRAM



Master SPI Quad (x4) configuration scheme

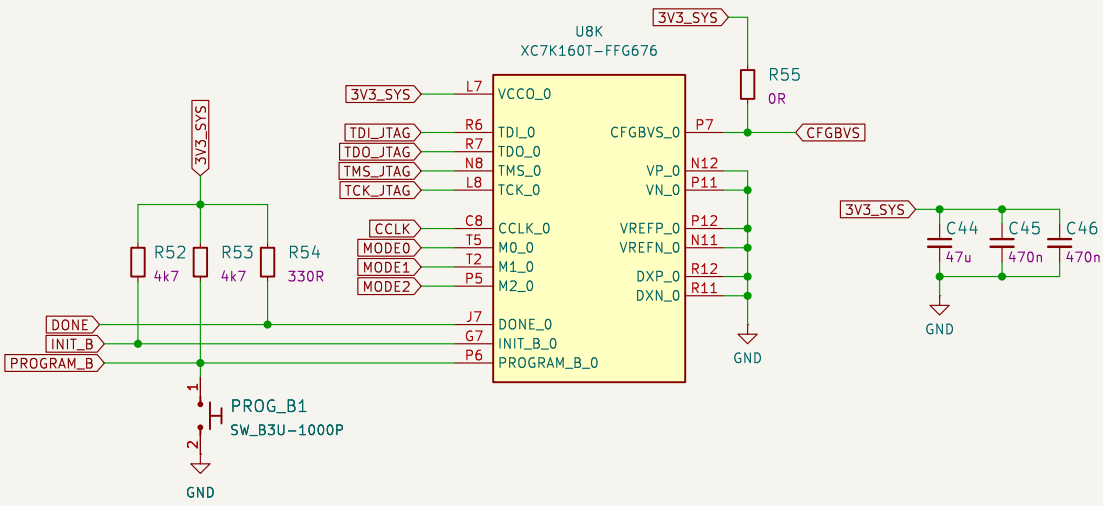
Follows Figure 2-14 7 Series FPGAs Configuration User Guide
UG470 (v1.13.1)

(Q)SPI flash



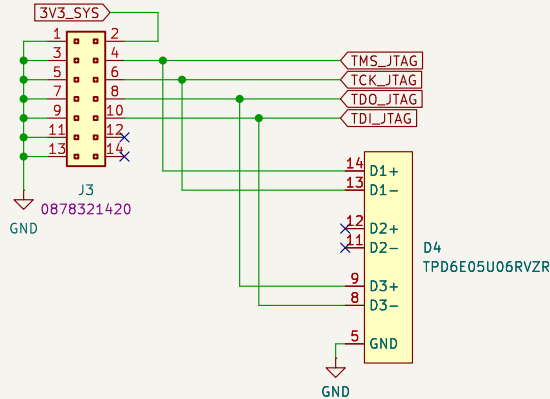
FPGA BANK 0

TODD: verify after FPGA swap

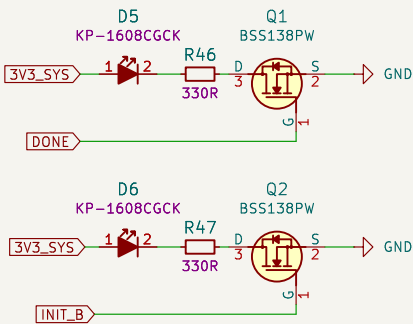


JTAG Connector

Compatible with Xilinx Platform Cable

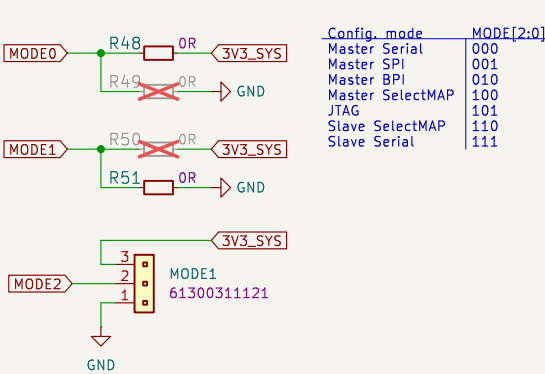


STATUS LEDs

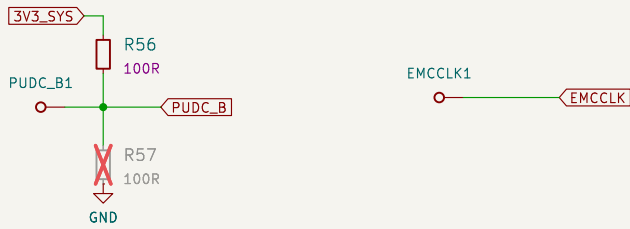


Configuration Modes

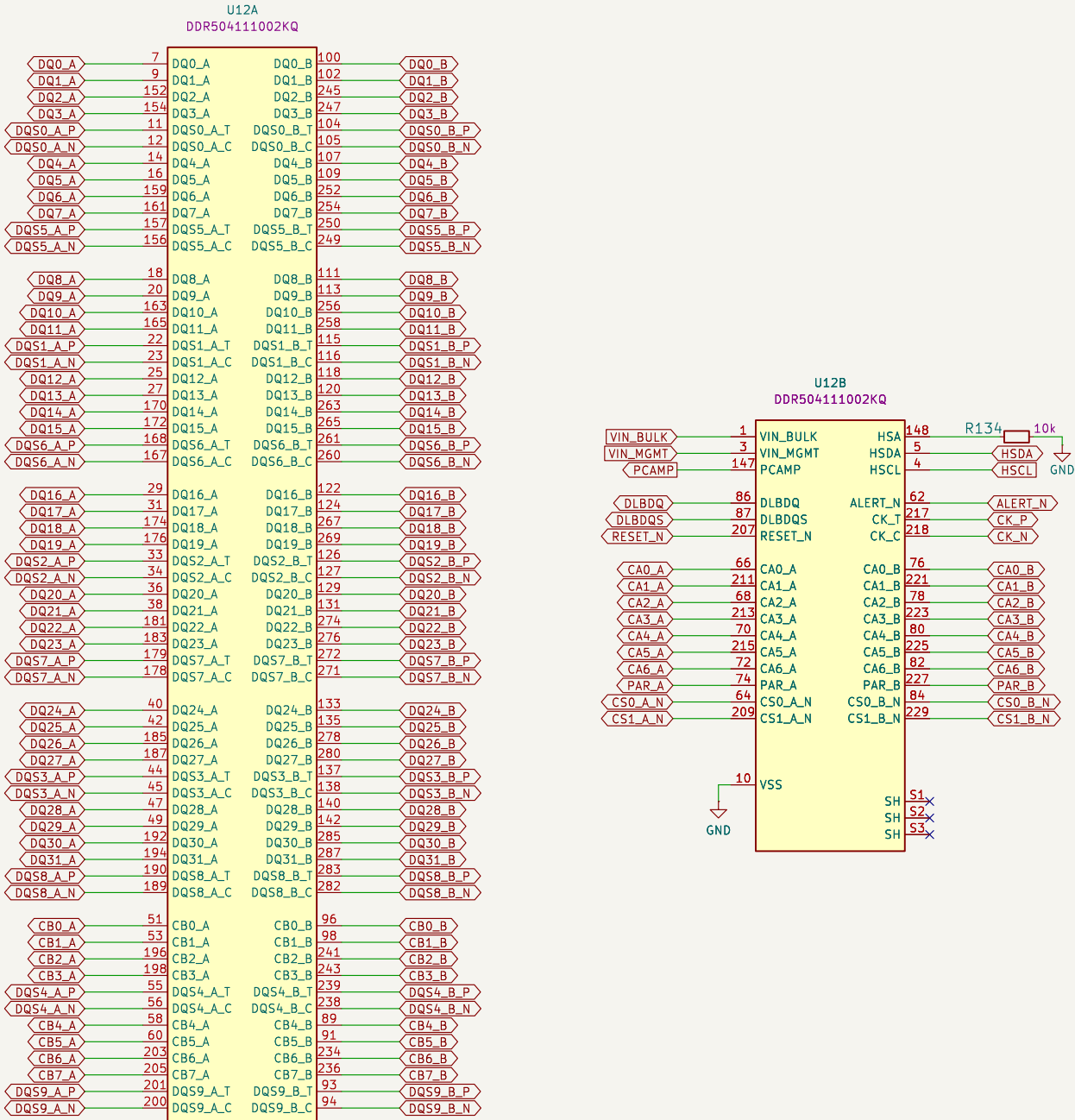
For details, see UG470 p. 21

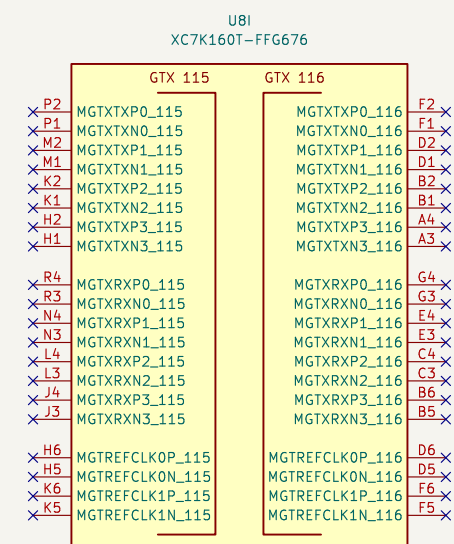
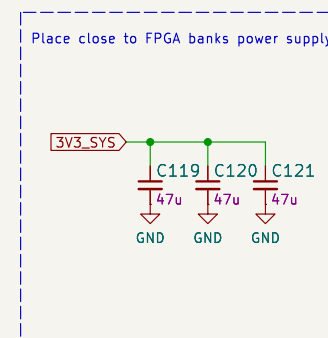
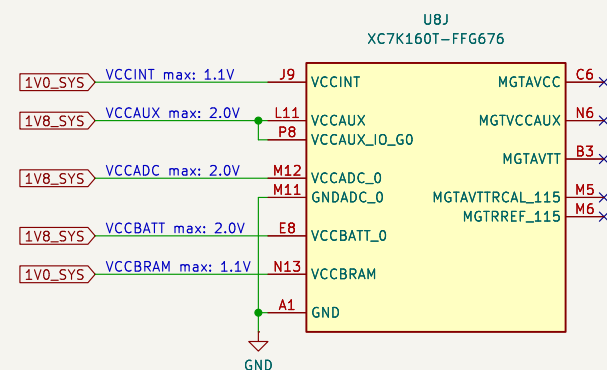
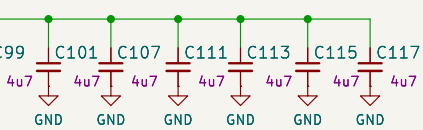
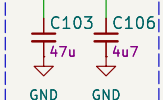
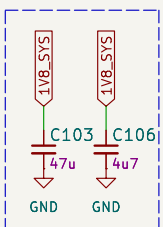


Probes



DDR5 RDIMM connector

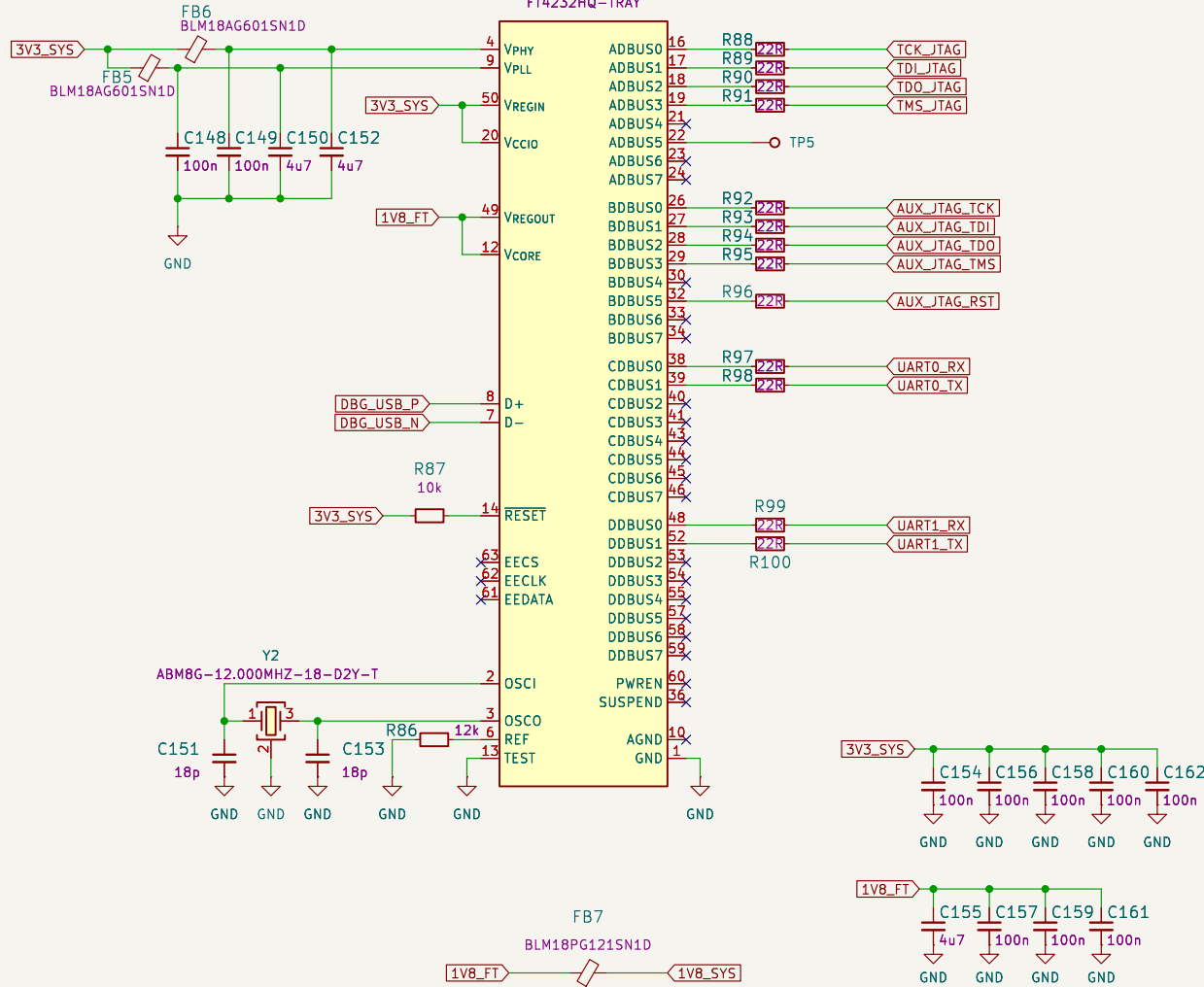




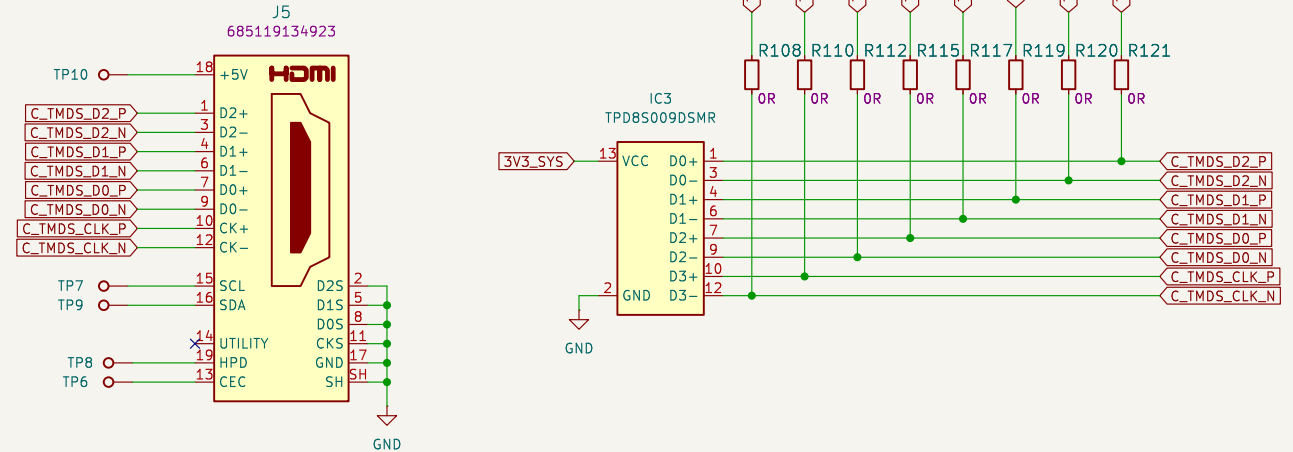
Debug UART

U10

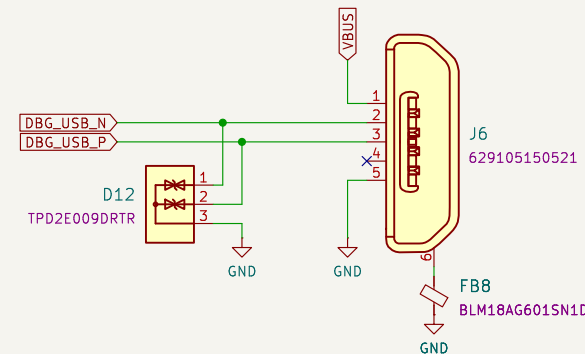
FT4232HQ-TRAY



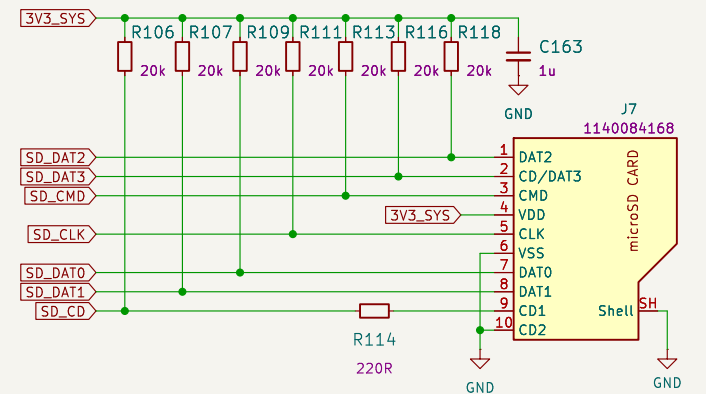
HDMI



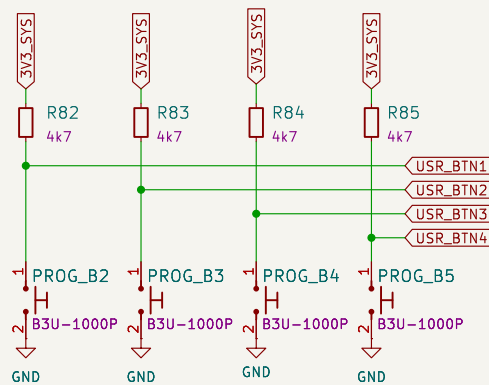
Debug USB



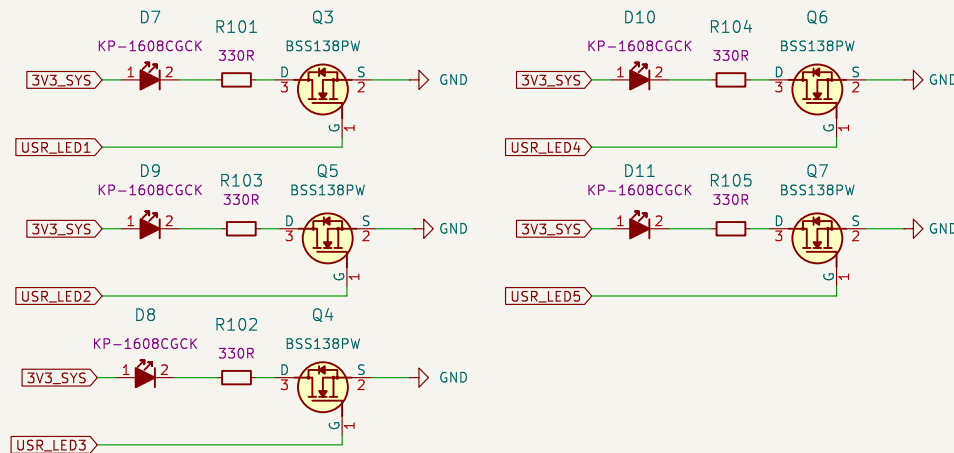
SD card

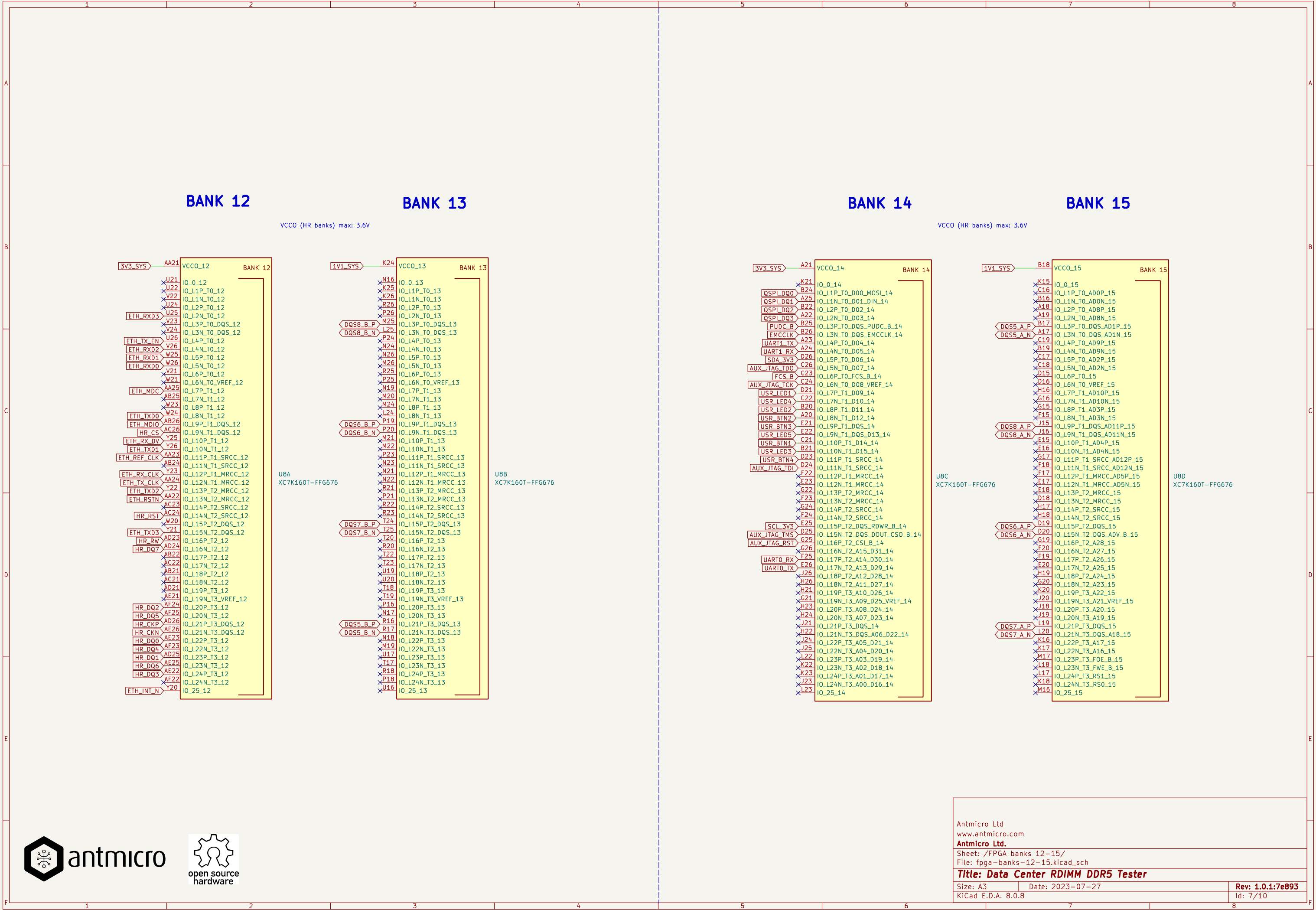


User buttons

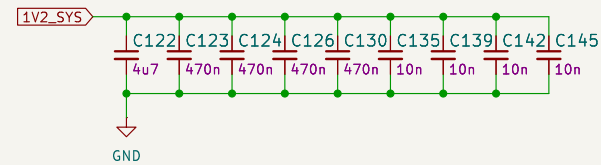


User LEDs

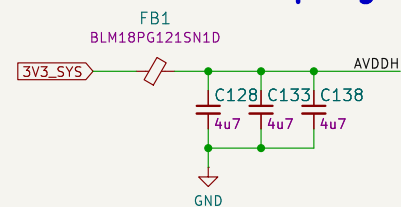




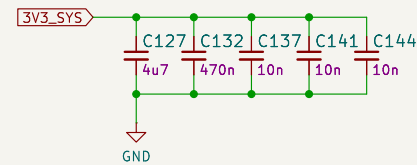
DVDDL decoupling



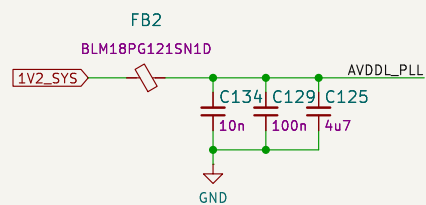
AVDDH decoupling



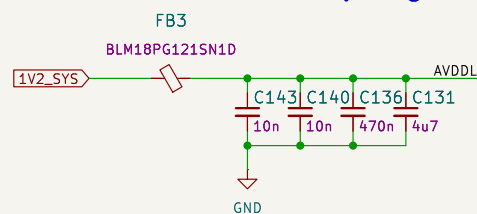
DVDDH decoupling



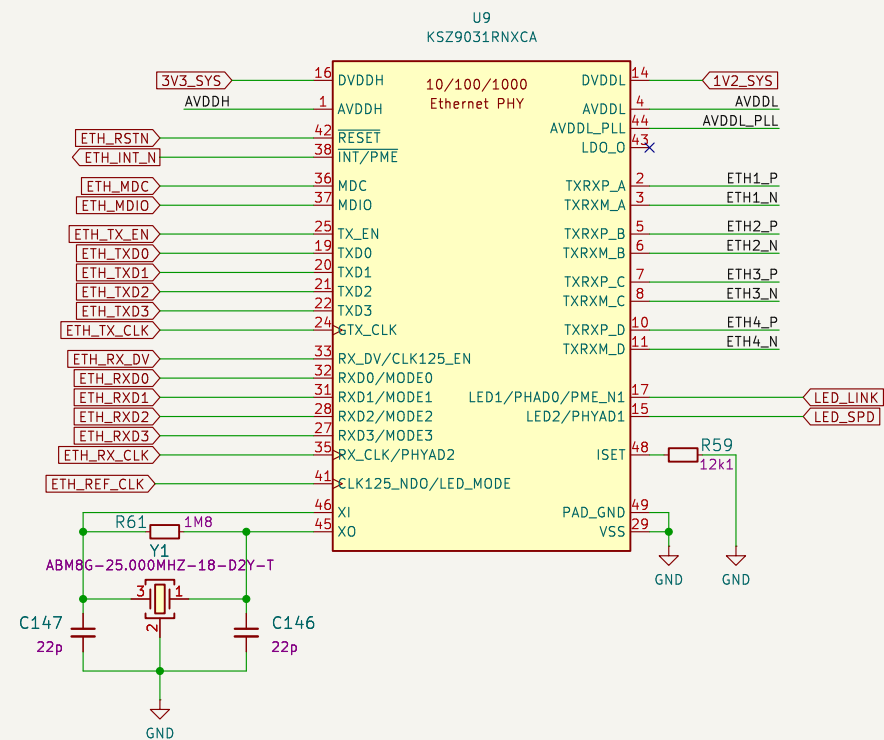
AVDDL_PLL decoupling



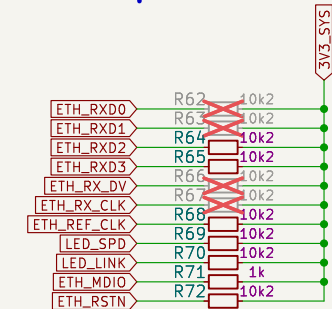
AVDDL decoupling



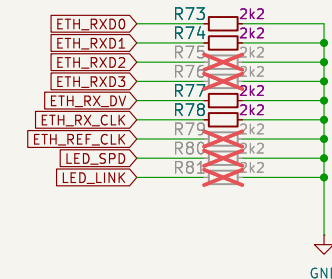
PHY



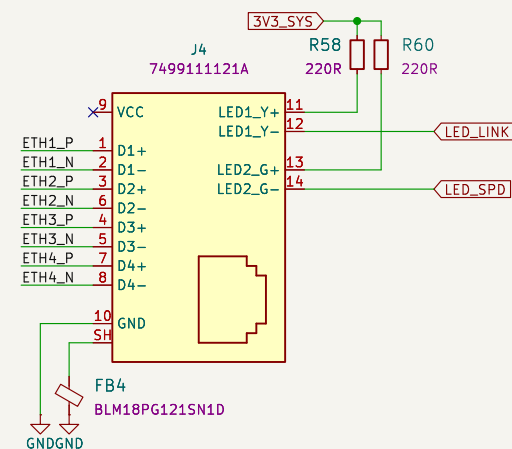
Pull up resistors

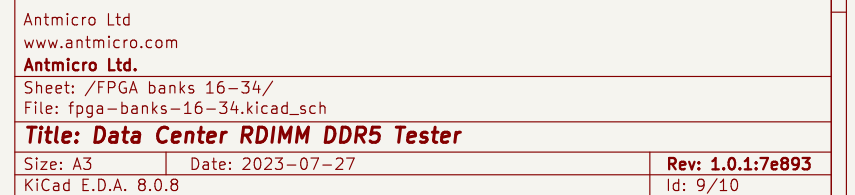
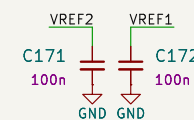
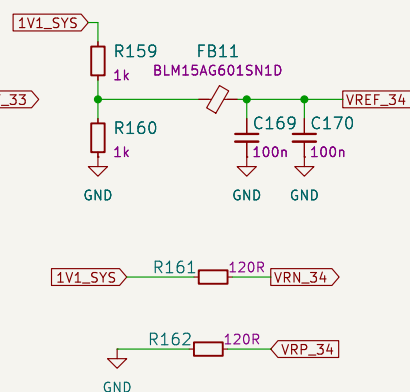


Pull down resistors



RJ45 Connector

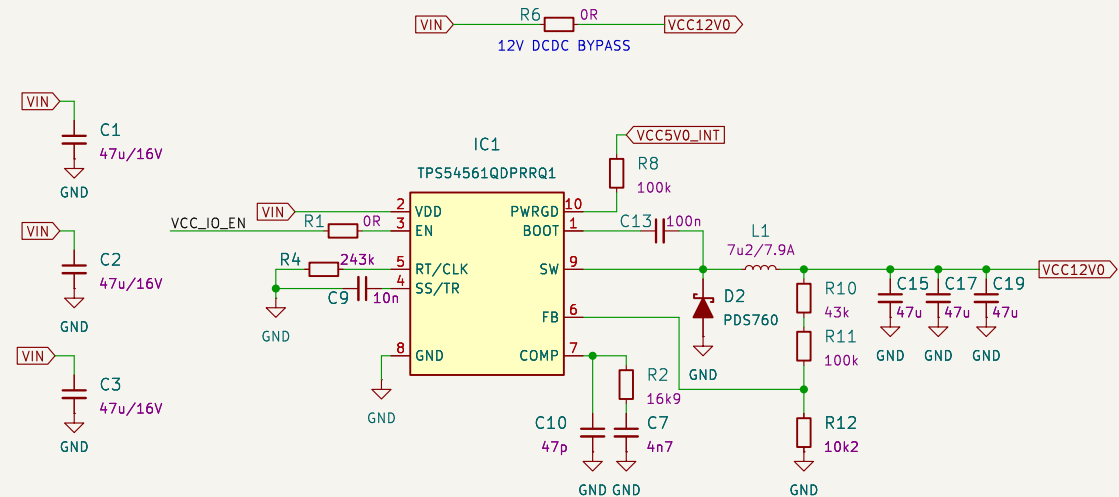




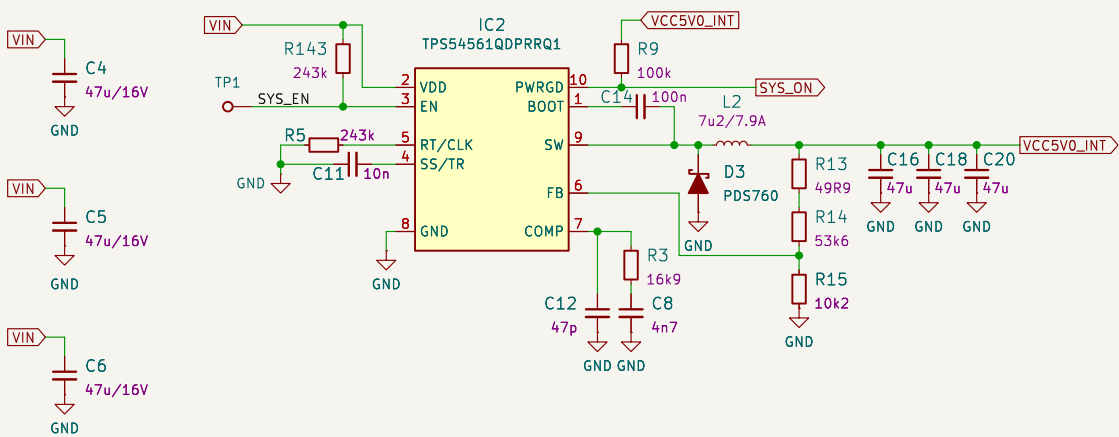
Input power connector



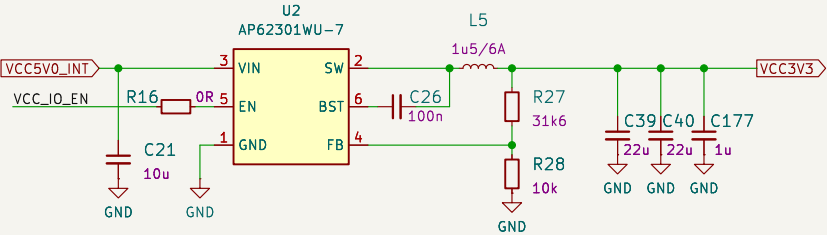
Main supply (12V 5A)



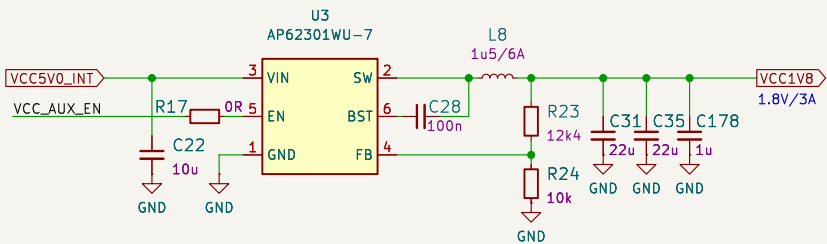
5V0 supply



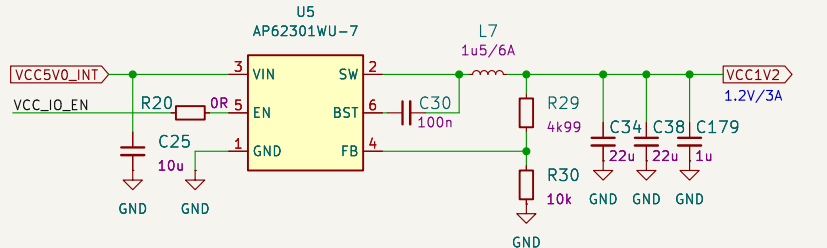
3V3 supply (3A)



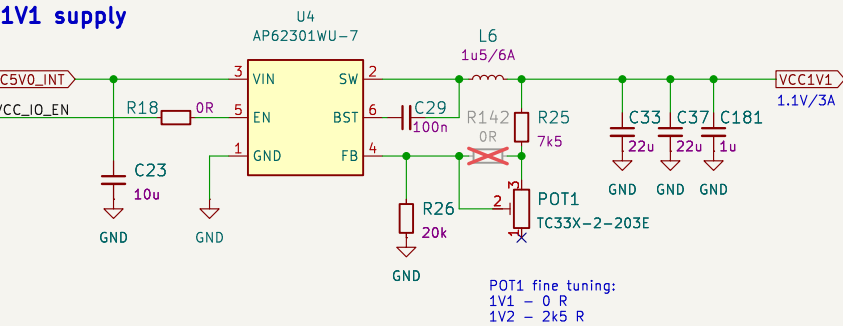
1V8 supply



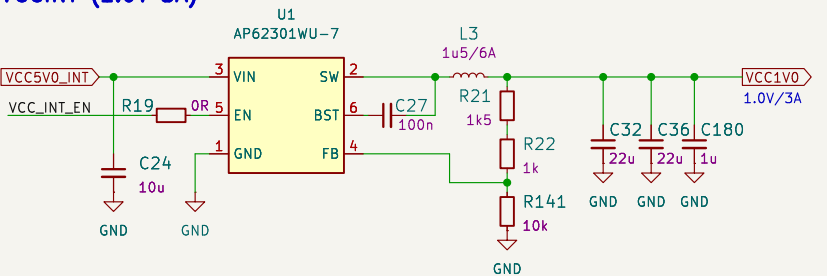
1V2 supply



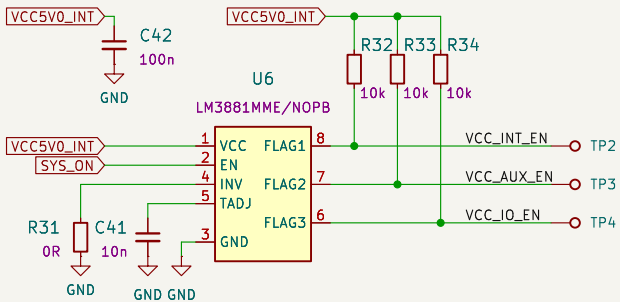
1V1 supply



VCCINT (1.0V 3A)

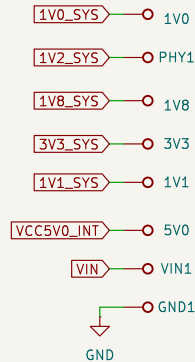


Power sequencer

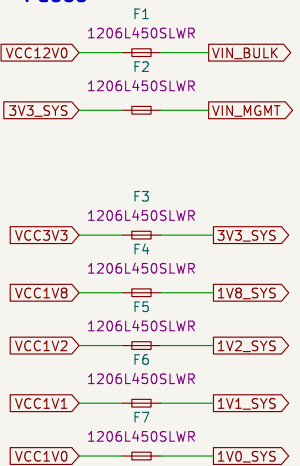


STEP1 - VCCINT (1.0V) for FPGA
STEP2 - VCCAUX (1.8V, 2.5V, 1.2V) for FPGA and DDR
STEP3 - VCCIO (3.3V, 1.2V, 0.6V) for FPGA, PHY and DDR

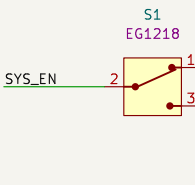
Probes



Fuses



Power switch



Optional FAN connector

